

2020 AZA CONSERVATION GRANTS FUND

Application Instructions

DEADLINE

This 2020 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT on 16 March 2020**. An auto-response email with a time-stamp will confirm receipt of the application.

*Incomplete applications, or applications received after 5:00 p.m. EDT on 16 March 2020, **regardless of cause**, will NOT be eligible for consideration. **NO DEADLINE EXCEPTIONS WILL BE MADE.***

MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit <https://www.aza.org/individual-membership>.

FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at <https://www.aza.org/cgf-selection-process>. Applicants will be notified of funding decisions in early September 2020 and funds will be available 1 October 2020. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

APPLICATION GUIDELINES

The CGF application and all supplemental forms are posted online at: <http://www.aza.org/cgf-information-for-applicants/>. These documents contain form fields that must be completed and submitted via email to: CGFapp@aza.org. *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. *Only* completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

APPLICATION INSTRUCTIONS AND CHECKLIST: All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

CGF Application: The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled “**Application**” and attached to your CGF Application email.

Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. NO EXCEPTIONS WILL BE MADE.

Note that “spell checker” and “track changes” functions cannot be used in this form. AZA strongly recommends that applicants compose the content for their Project Abstract and Project Proposal in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal.*

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list. Include a project abstract no longer than 1,500 characters, including spaces.
- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write “N/A” in the field but *do not leave any form fields blank*, as this may delay processing of your application.
- ☒ **Budget.** Required budgetary information includes:
 - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will not be covered by CGF funds.
 - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

Required Attachments: The following must be attached to your CGF Application email:

- ☒ **Curricula Vitae.** A curriculum vitae (CV) is required for **each** PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed a **maximum** length of two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, “**CV Investigator’s Last Name.**” *CVs that exceed the maximum length will be cut off at the second page.*
- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI’s institution and **EACH** CI institution *and* collaborating institution must be attached to the CGF Application email as separate documents titled “**SIS**

Institution's Name.” The *Statement of Institutional Support* form can be found with the rest of the CGF Application materials on the AZA website identified above.

Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions, or AZA Animal Programs may submit such items as appendices.

Note that letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form. In addition, letters of support/endorsement **from AZA Animal (Taxon Advisory Groups, Species Survival Plans®, etc.) and SAFE Programs** are not accepted although coordination with AZA Animal and SAFE Programs is encouraged and should be described in Question 1. To assure Animal and SAFE Program review of all relevant CGF submissions, AZA Program Leaders will serve as expert reviewers in the CGF application review process and may express their support for a proposal at that time.

- ☒ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your CGF Application email. Each permit attached to the CGF Application email must be titled “**Permit #.**” *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2020 to ensure award eligibility.*
- ☐ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for CGF funding and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the CGF Application email. This attachment must be submitted as a Microsoft Word document titled “**Resubmission Form**” and attached to your CGF Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the CGF Application materials on the AZA website identified above.

Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.

Non-Required Attachments: The following attachments to your CGF Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Project Proposal should be included in a separate document titled “**Citations**” and attached to the CGF Application email.
- ☒ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos or publications. Appendices must be attached to the CGF Application email titled “**Appendix #.**”

Note: Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

SUBMISSION INSTRUCTIONS

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA’s website at <http://www.aza.org/cgf-information-for-applicants/>.

Proposals must be submitted electronically using the 2020 CGF Application and related forms via a **single email, with the proposal title as the subject line**, to CGFapp@aza.org no later than 5:00 p.m. EDT on 16 March 2020. Hard copies of any application materials or applications received after this deadline will not be accepted *regardless of cause*.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2020 CGF Application email **must** include the following attachments

- CGF Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution; and
- A Proposal Resubmission form, if applicable.
- The Application email may also include citation and appendices attachments.

Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, s/he should ensure that an auto-response was received for *each* submission. If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the CGF Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

APPLICATION ASSISTANCE

The CGF website (<http://www.aza.org/cgf-information-for-applicants/>) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 7 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

2020 AZA CONSERVATION GRANTS FUND
Cover Sheet

Project Number: 20-1681
For AZA use only

Title of Project: Intersecting human land-use, livelihoods, and conservation of the endangered Zanzibar red colobus

Amount Requested: \$30,878 (round to the nearest whole dollar)

Project Start Date: January 2021

End Date: December 2021

(CGF funding is for one year only, beginning no earlier than the award date)

Principal Investigators: Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

1. Principal Investigator (PI) Name and Title: Jenneca McCarter, MSc

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (required; available at: <http://www.aza.org/cgf-information-for-applicants/>) ☒

Individual AZA Member Number: 81087768 (required)

Institution/Organization: University of Oregon

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: University of Oregon, Department of Anthropology, 302D Condon Hall, 1321 Kincaid St, Eugene, OR, 97403

Phone: 9182894453

E-mail: jennecam@uoregon.edu

2. Co-Investigator (CI) Name and Title:

CV Attached? ☐ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (available at: <http://www.aza.org/cgf-information-for-applicants/>) ☐ (required)

Individual AZA Member Number: (required)

Institution/Organization:

Statement of Institutional Support Form Attached? ☐ (required)

Institutional Address:

Phone:

E-mail:

3. Co-Investigator (CI) Name and Title:

CV Attached? ☐ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions (available at: <http://www.aza.org/cgf-information-for-applicants/>) ☐ (required)

Individual AZA Member Number: (required)

Institution/Organization:

Statement of Institutional Support Form Attached? ☐ (required)

Institutional Address:

Phone:

E-mail:

Note: If additional Co-Investigators need to be included, add their information in a separate document titled “**Additional Co-Investigators**” and include it as an attachment to your CGF Application email.

Collaborating Institutions or Organizations: A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the “Grant/Matching Support” section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the PI’s and CI’s institutions listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the CGF Application email.

Collaborating Institution Names:

Bangor University *Statement of Institutional Support* Form Attached? ☒ (required)

Wildlife Conservation Society (Tanzania) *Statement of Institutional Support* Form Attached? ☒ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled “**Additional Collaborators**” and include it as an attachment to your CGF Application email.

Project Categories (choose two that best describe the project):

☐ Animal/Population Management (*Ex situ*)

☒ Research

☐ Conservation Education

☐ Animal Health

☒ Field Conservation (*In situ*)

☐ Animal Welfare

Project Abstract (limit 1,500 characters with spaces):

Red colobus monkeys are among the most threatened groups of primates in the world, with more than half of the taxa classified as either Endangered or Critically Endangered on the IUCN Red List of Threatened Species. One of these monkeys is the endangered Zanzibar red colobus (ZRC), an endemic species found in the Zanzibar Archipelago, part of the United Republic of Tanzania. ZRC are facing an increasingly uncertain future due to the combination of habitat loss, habitat fragmentation, hunting, and the environmental impacts of climate change. This project will address critical needs for ZRC conservation by producing ZRC genetic data and human behavioral data, which will then be combined to directly inform conservation management recommendations. By utilizing a landscape genetics approach, our data will allow us to determine how ZRC migration across the landscape is affected by different anthropogenic landscape features. The human data will illuminate determinants of human land-use and flexibility of livelihoods in ZRC areas, particularly within the framework of climate change, both as it is being experienced in the present and how it may alter livelihoods and land-use patterns in the future. These two results will then be combined to create a spatially explicit recommendation for land-use planning that maintains ZRC genetic variation and adaptive potential while also strengthening resilience in human livelihoods to climate change.

2020 AZA CONSERVATION GRANTS FUND

Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance for AZA animal/conservation/education programs or institutions, and/or how it reflects the priorities of the AZA Board (e.g., sustainability of managed populations, SAFE: Savings Animals From Extinction) or of related Animal or SAFE species programs. **LIMIT 5,000 CHARACTERS with spaces**

The Zanzibar red colobus monkey (ZRC) is an endangered species endemic to the Zanzibar Archipelago (United Republic of Tanzania), which is part of the 'Coastal Forests of Eastern Africa' biodiversity hotspot [1]. The ZRC is native to Unguja Island (1,670 km²) and has a current population size of 5,862 individuals, which are distributed across roughly 21 sites in a fragmented manner [2]. ZRC are facing an increasingly uncertain future due to a combination of habitat loss, habitat fragmentation, hunting, and climate change [3]. A long history of economic reliance on agriculture, a recent sharp reliance on tourism, and steep human population growth has led to intense resource use and land conversion, with rates of deforestation on Zanzibar reaching ~10 km² per annum [4, 5]. This has left the ZRC to navigate a highly heterogeneous landscape with a suboptimal agricultural matrix between its preferred forest habitats. Approximately 31% of the total population exists outside of protected areas and some of these sites are very isolated. We currently lack knowledge of how landscape discontinuities impact ZRC dispersal between forest patches and thus gene flow, which is crucial for the maintenance of genetic diversity in populations. Genetic diversity can act as a buffer against stochastic processes, and populations with low genetic diversity have been shown to be more at risk to extinction via disease outbreak and environmental change [6, 7, 8]. Understanding the current dispersal patterns of ZRC is thus crucial to maintaining adaptive potential and building effective population management.

Informed population management of the ZRC also requires incorporation of climate change. Several studies have established that climate change has already caused quantifiable shifts in sea levels, patterns of precipitation, and soil quality on Unguja [9, 10, 11]. The ZRC will not only be directly impacted by these environmental and phenological shifts as a forest dependent species with a folivorous diet [12], but they will also face indirect pressure due to the effects of climate change on human livelihoods in ZRC areas. This is especially problematic as poverty is a persistent issue in Zanzibar, with 30% of people living on less than \$1.90 USD a day. Economic instability creates a strong reliance on the land and natural resources, while environmental changes create uncertainty in these practices. These issues likely feedback on one another, resulting in highly vulnerable human livelihoods and an increased level of dependence on the land. ZRC share habitats and resources with humans across their range, meaning their long-term viability as a population is also reliant on humans having stable livelihoods. This makes understanding the determinants of human livelihoods and how resilient they are to climate change an important element of creating effective conservation strategies for ZRC.

The overarching goal of this research is to support long-term survival of the ZRC population, while also building resilience to climate change into human livelihoods. The specific questions addressed will be: (1) how do specific types of land-use impact gene flow in ZRC? (2) what are the determinants of different types of human land-use near ZRC habitats and how might these change in the future? and (3) how flexible are people's livelihoods in ZRC habitat areas and can changes be made to enhance ZRC migration? This project will combine spatially explicit ZRC biological data and human behavioral data to create management recommendations that prioritize the most viable ZRC subpopulations for conservation while simultaneously suggesting land-use schemes which encourage ZRC migration and resilience in human livelihoods. Such land-use planning is a critical need for the future survival of the ZRC and the well-being of people on Zanzibar.

Unguja is extremely vulnerable to climate change as a small low lying island state, tying this research directly to AZA's climate change position statement. AZA also prioritizes research related to species and habitat conservation. The ZRC is an endangered species endemic to a biodiversity hotspot, as well as the flagship species for Zanzibar and central to ecotourism critical to sustainable development plans aimed at poverty relief. Lastly, the AZA CGF aims to empower zoo professionals to be leaders in their field. The PI has nearly a decade of experience working in AZA accredited facilities, including five-years as a primate zookeeper at the Tulsa Zoo. This provided the PI with experience directly applicable to the proposed project, but more importantly inspired her to pursue her doctorate and actively contribute to primate conservation. The PI strongly believes that ex-situ and in-situ partnerships strengthen conservation initiatives, and would

like to one day take her personal knowledge and experiences back to the AZA community through employment after completion of her doctorate.

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

The long-term goal of this research is to ensure the survival and viability of the ZRC. This directly relates to the goal of the proposed project, which is the creation of a spatially explicit land-use plan that simultaneously ensures the long-term viability of the ZRC and supports human livelihoods by building resilience to climate change. This will be achieved by understanding how land-use influences human livelihoods and alters patterns of ZRC movement, which will then allow for landscape planning which takes both into account. At the end of the research, management recommendations will be distributed to the appropriate governmental and public stakeholders in order to support local decision making. To accomplish the project goal, the following objectives will be completed:

Objective 1: Determine how land-use on Unguja impacts movement and migration of ZRC through a landscape genetics approach. This will allow for the identification of landscape features that influence genetic connectivity, including which specific land-use types and crops are more permeable to ZRC dispersal. In order to complete this objective, 1) non-invasive (i.e. fecal) georeferenced samples will be collected across forests containing ZRC groups (~21 sites, ~300 samples), 2) samples will be genotyped, 3) GIS data will be collected for the island, 4) population genetic analyses will be conducted to identify how ZRC genetic variation is structured across the island, 5) landscape genetics analyses will be conducted to identify which landscape features impede ZRC movement and which ones allow for ZRC gene flow.

Objective 2: Identify the determinants of human land use in ZRC areas, including how flexible people are in their agricultural practices.

Semi-structured interviews will be conducted to generate information concerning agricultural practices and land tenure, human-wildlife interactions, socioeconomic factors, and climate change. In order to complete this objective, (1) potential participants will be identified from red colobus habitat adjacent communities through stratified selection using land-use types and distance to natural forest patches as proxies for potential human-colobus interaction, (2) key informant interviews and quantitative household surveys will be carried out using Participant Rural Appraisal tools, (3) transcripts from interviews will be analyzed by identifying common themes and collating related statements, (4) household surveys and appropriate data from interviews will be coded as binary, ordinal, and continuous data, and (5) spatially explicit social data models will be created. Knowing how and why people are using the land will enable us to recommend land-use schemes that encourage ZRC migration, are politically and socially tenable, and build resilience in the human community to climate change.

Objective 3: Creation of spatially explicit management recommendations for ZRC.

The results from Objectives 1 and 2 will be combined to create conservation recommendations which maintain adaptive potential and genetic diversity for the ZRC and are also socially and politically tenable. In order to meet this objective we will, (1) employ geospatial software to overlay the landscape genetics and social data, (2) utilize geospatial analysis techniques, such as network analysis, to assist in prioritizing regions for conservation action, (3) once regions are selected, we will revisit the original data in order to determine what type of land-use scheme (i.e., shifts in agricultural practices or crop types) best suit those specific regions, and (4) once recommendations have been finalized, they will be compiled into a document for dissemination.

This project has an estimated total timeline of 15 months. The portion of the project a CGF grant would fund is the first 12 months of this period. This includes a field season followed by lab work, analyses, and creation of conservation recommendations. Dissemination of results will occur during the three month period after CGF support ends. The outputs from the 12 months of work funded by a CGF grant would include all of the results from the analyzed ZRC genetic and human social data as well as the conservation recommendations. In the long-term, the conservation management recommendations created by this project will strengthen viability of the ZRC population across its range.

3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and proposed actions are appropriate for addressing the conservation issue addressed. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000 CHARACTERS with spaces**

In order to address the question (1) how do specific types of land-use impact gene flow in ZRC, a landscape genetics approach will be taken. This will utilize standard methodology for sample collection, microsatellite genotyping, and analysis [13]. Non-invasive fecal samples will be collected in the field and stored in tubes with 6 mL of RNAlater in order to stabilize DNA. Accompanying GPS tags will be marked for each sample. The PI will accomplish this by following ZRC groups at a distance; there will be no direct contact with any animals during sampling and personal protective equipment will be worn to protect both the PI and the animals. The sampling scheme is based on a population census completed in 2015, which estimates there are 342 social groups at roughly 21 sites across the island [2]. 300 samples will be collected across the island, roughly one sample per social group, in order to create a continuous sampling distribution [14, 15]. The PI will attempt to sample as many different social groups as possible, but in the event not all social groups can be located, locales with a higher density of social groups will be sampled more intensely. The PI has experience carrying out fieldwork of this exact setup with wild chacma baboons (*Papio ursinus*) in South Africa [16], and collaborators Dr. Alex Georgiev and Zoe Melvin have been collecting ZRC fecal samples for hormonal analysis since 2018, showing the feasibility of this approach. A permit is required from the Revolutionary Government of Zanzibar's Department of Forestry and Non-Renewable Natural Resources to collect samples from the colobus. Collaborator Dr. Alex Georgiev will aid in obtaining permits and has successfully secured permits for non-invasive colobus sampling in the past.

Samples will be exported to the United States for DNA extraction and analysis. Sample preparation and DNA extraction will be conducted in the Ting Lab at the University of Oregon. This project will use well-tested and vetted methodology to genotype microsatellites from fecal-extracted DNA. We will genotype each sample at 15 microsatellite loci, as microsatellites are highly variable and it has been shown that as few as 8-12 loci can be sufficient to distinguish related individuals with statistical confidence in a landscape genetics study [17]. Previously developed microsatellites are available for colobus monkeys, which enables us to carry out this research without needing to develop our own markers [18]. In order to bring the costs of the total lab analysis down, we will use multiplex PCR to reduce the overall number of reactions that need to be carried out. The samples will then be genotyped using an ABI 3730 capillary DNA sequencer at Oregon State University's Center for Genome Research and Biocomputing. Multi-locus genotypes will be combined with remotely sensed geospatial data for Unguja. Geospatial analysis software, such as ArcMap, will be utilized for these analyses. For this project, isolation-by-distance (IBD) is the null hypothesis. IBD assumes that genetic difference is best explained by geographic distance. Our operating hypothesis is that genetic difference will be better explained by anthropogenically created landscape features than it can be by distance alone. In order to test this hypothesis we will use the isolation-by-resistance (IBR) model and circuit theory [19]. Isolation-by-resistance predicts a positive relationship between genetic and resistance distances, a theoretic distance metric based on circuit theory. Circuit theory processes landscapes as if they were conductive surfaces, with electrons flowing across a circuit to model genes moving across the landscape [20]. Circuit theory allows one to consider all possible paths between two samples when assessing landscape influence on gene flow. Circuit-based connectivity will be calculated using the software Circuitscape 4.0. In order to do this, we will first develop resistance surfaces for the different landscape features we want to test as barriers to ZRC movement across the landscape. Resistance surfaces will be created utilizing remotely sensed data and will be parameterized using the R package ResistanceGA [21]. The output from this process will be a model of ZRC gene flow across the island, which features hinder ZRC gene flow, and which anthropogenic land-uses or crop types are most permeable or impermeable to ZRC dispersal.

In order to address the questions (2) what are the determinants of different types of human land-use near ZRC habitats and how might these change in the future, and (3) how flexible are people's livelihoods in ZRC habitat areas and can changes be made to enhance ZRC migration, we will employ social science methodologies. Many lines of questioning will be nested within these broader questions in order to understand agricultural practices, land tenure, human-wildlife interactions, socioeconomic factors, and climate change. A mixed-methods approach will be carried out, combining informant interviews, which utilize Participatory Rural Appraisal tools [22], and quantitative household surveys. Contacts will be made and the PI will set up interviews in one-on-one and qualitative focus group settings. We are most interested in eliciting information from farmers, but we may decide there are other key informants, such as individuals working in non-governmental conservation organizations, important to interview. Participants will be identified from red colobus habitat adjacent communities through stratified selection based on land-use types and distance to natural forest patches as proxies for potential human-colobus interaction [23]. All information collected will be done so in a way that preserves the participants' anonymity. The PI has intermediate proficiency in Swahili and is continuing to develop her language skills through coursework at the University of Oregon so that she can perform the interviews herself.

Transcripts from interviews will be analyzed by identifying common themes and collating related statements [24]. Household surveys, which comprise our socioeconomic data, will be coded as binary, ordinal, and continuous data as appropriate. We will also quantitatively code any appropriate data from our interviews, such as categorical coding of crop types or binary coding of subsistence versus commercial agriculture [25]. This will allow us to create spatially explicit social data models. We will take this information and utilize geospatial processing software, such as ArcMap, to create raster and vector models which display continuously and discretely distributed data across the landscape. Once we have these data layers, we will be able to overlay the social data layers onto the landscape genetics analysis, producing a

biologically and socially informed spatially explicit model. This will allow us to create spatially explicit recommendations for land-use planning that incorporates the needs of both the ZRC and human livelihoods.

4. If previous CGF funding was obtained for this or a related project, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

N/A

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

This entirety of this project will take roughly 15 months to complete, beginning with fieldwork and ending with the dissemination of findings and recommendations to stakeholders in Zanzibar. The one year portion supported by a CGF grant would include field work, lab work, and data analysis and is detailed below.

- January 2021 – July 2021 (6 months): The field season will be focused on non-invasive genetic sampling across Unguja with the help of field assistants, as well as conducting semi-structure interviews with farmers living in colobus habitat areas.

- July 2021 - December 2021 (6 months): Lab work to genotype samples from the 2nd field season. Landscape genetics analyses will then be completed utilizing the genetic data and geospatial data. Data from the semi-structured interviews will be analyzed utilizing mixed-methods. Conservation recommendations will be created by combining the ZRC genetic and human social data.

Additionally, we are very aware of the uncertainty that the current COVID-19 outbreak presents. The timeline of this project has already been shifted due to current restrictions in travel, if however there are still restrictions on travel come winter 2021, we imagine our entire timeline could be shifted by 3-6 months, with the proposed work pushed back to a summer 2021 start date.

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

Jenneca McCarter (PI) is a PhD student at the U. of Oregon, and the proposed project represents her doctoral research. McCarter has her MSc in Conservation Biology from the University of Cape Town, where her research focused on using immunogenetics to inform conservation management of chacma baboons. McCarter has a wide variety of field and lab experience, as well as over a decade of experience working at AZA accredited institutions, giving her a broad perspective of in ex-situ and in-situ primate conservation. McCarter will be the researcher collecting all of the data in the field, will do all of the lab work and data analysis, and will create the subsequent conservation recommendations and disseminate them.

Nelson Ting, PhD (Collaborator) is an Associate Professor at the U. of Oregon and specializes in the use of molecular genetics to aid in the conservation of endangered species. His research has focused on colobus monkeys, and he is a lead editor of the Red Colobus Conservation Action Plan (ReCAP). Dr. Ting aided in the development of the proposed research and will oversee all aspects of the project including data collection in the field, analyzing and interpreting results, and the creation and dissemination of conservation recommendations.

Dr. Alex Georgiev (Collaborator) is a Lecturer at Bangor U. and Director of the ZRC Project (ZRCF), which studies how the endangered ZRC is coping with human-induced changes to their habitat. Dr. Georgiev has provided feedback on project design and will assist with on-the-ground research logistics in Zanzibar, as well as the creation of conservation recommendations based on these data.

Dr. Tim Davenport (Collaborator) is the WCS Director for Species Conservation in Africa and the former WCS Country Director for Tanzania. He has worked across Africa and is involved in research, protected area design and management, climate change adaptation, and community conservation initiatives. Dr. Davenport has provided essential feedback on project design, will be a source for on-the-ground logistics in Zanzibar, and will also assist with the creation of conservation recommendations and communications with the Zanzibar government.

Zoe Melvin (Collaborator) is a PhD student at Bangor U. She has studied ZRC groups in multiple locations across Unguja since 2018, validating non-invasive collection of fecal samples. Melvin has provided feedback on project design and will assist with sample collection and logistics in the field.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed). **LIMIT 3,000 CHARACTERS with spaces**

We will monitor the progress of the proposed research over the course of the 12 month funding period. We will use a process indicator system, making sure actions are complete in a timely manner (see proposed timeline). The actions we will monitor for progress and completion in each objective are:

Objective 1: Determine how land-use on Unguja impacts movement and migration of ZRC through a landscape genetics approach. 1) non-invasive (i.e. fecal) georeferenced sample collection across forests containing ZRC groups (~21 sites, ~300 samples), 2) sample genotyping 3) GIS data collection, 4) population genetic analyses 5) landscape genetics analyses

Objective 2: Identify the determinants of human land use in ZRC areas, including how flexible people are in their agricultural practices. (1) identification of potential participants from red colobus habitat adjacent communities, (2) completion of key informant interviews and quantitative household surveys using Participant Rural Appraisal, (3) analysis of transcripts from interviews (4) coding of household surveys and appropriate data from interviews as binary, ordinal, and continuous data, and (5) creation of spatially explicit social data rasters.

Objective 3: Creation of spatially explicit management recommendations for ZRC. (1) overlay of landscape genetics output with social data models in geospatial analysis software, (2) determination of which regions should be prioritized for conservation action in order to support dispersal of red colobus, (3) determination of which areas human livelihoods are most vulnerable, (4) creation of recommendations that combine this information, and (5) delivery of recommendations to stakeholders.

We will evaluate the impact of the research following project completion for at least 2 years. Specifically, we will identify which of the recommendations we make are converted to policy, how that policy is implemented, whether it encourages ZRC movement across the landscape, and how it is received by people living adjacent to ZRC habitat.

8. Explain and identify the means, and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

The Red Colobus Conservation Action Plan under the IUCN/SSC Primate Specialist Group calls for the integration of local communities into red colobus conservation through bottom up approaches, and for the identification and prioritization of ZRC conservation units in order to strengthen conservation in and around protected areas. Completion of the proposed research would assist in simultaneously addressing these objectives by identifying requirements necessary to ensure a sustainable future for humans and the ZRC. The results from the proposed work will be disseminated directly by the PI and collaborators to relevant stakeholders (e.g., Department of Forestry and Non-Renewable Natural resources, communities in ZRC habitat areas) in the 3 month period after CGF support ends, as well as via publication in peer-reviewed journals, presentations at conferences, and as a doctoral dissertation. Written reports will be given to the IUCN/SSC Primate Specialist Group, as the IUCN 'research needed' section calls for data on ZRC threats, area-based management plans, population trends, and habitat trends. The PI will also engage in science communication throughout the project, utilizing social media platforms to share information and research experiences with the general public. The PI believes making these results available and approachable for both local and international stakeholders is key to engendering support not just for ZRC conservation, but for wildlife conservation in general.

9. Have all the necessary permits been obtained for the project?

a) ☐ Yes ☒ No ☐ Not Applicable

If yes, list the permits here and attach one copy of each to your CGF Application email. Attachments should be titled "**Permit #.**"

N/A

- b) If the necessary permits have not yet been obtained, please explain why. Pending permits must be obtained and submitted to AZA by 15 June 2020.

We have not yet acquired permits as fieldwork is not starting until January 2021. Our research group, the Zanzibar Red Colobus Project (ZRCF), has successfully secured permits for non-invasive Zanzibar red colobus sampling in the past.

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

N/A

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

N/A

12. **Past CGF Support.** If any of the proposal's Principal or Co-Investigators listed on the Cover Sheet have received previous CGF support, complete the required information below for each.

- | | | | |
|----|---|------------------------------|-----------------------------|
| a) | Name of Investigator: N/A | Year of Award: N/A | Amount of Award: N/A |
| | Project Title: N/A | | |
| | Were all required reports (annual/final) submitted? | ☐ Yes | ☐ No |
| b) | Name of Investigator: N/A | Year of Award: N/A | Amount of Award: N/A |
| | Project Title: N/A | | |
| | Were all required reports (annual/final) submitted? | ☐ Yes | ☐ No |

Note: If additional awards need to be included, add their information to a separate document titled "Past CGF Projects" and include it as an attachment with your CGF Application email.

2020 AZA CONSERVATION GRANTS FUND

Project Budget

BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x \$25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over \$500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

CGF does not fund the following: Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation/management/education goals.

CGF does not fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.

Note: Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's CGF Scientific Advisory Committee may recommend that the CGF award amount be reduced or refunded.

Note: If additional budget items need to be included, add this information in a separate document titled "**Additional Budget Items**" and include it as an attachment with your CGF Application email.

Budget Items Form:

Budget Item	Method of Calculation (e.g., 2 persons x 10 days x \$25/day)	CGF- Requested Amount	Non-CGF Requested Amount (if applicable)	Date Expected from Non-CGF Source (if applicable and not yet received)	Name of Non-CGF Source (e.g., Grant Fund, Matching)	Total Expense (CGF+Non-CGF)
Air Fare	1 person x 1 roundtrip ticket from Eugene, OR to Zanzibar	\$2,100	\$2,100	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$2,100
Visa	1 person x 1 visitor visa for United Republic of Tanzania	\$100	\$100	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$100
Field Assistants	3 field assistants x 100 days x \$15/day	\$0	\$4,500	May 2020	University of Oregon	\$4,500
Jozani-Chwaka Bay National Park fee	1 person x 1 park entry fee for six months	\$200	\$200	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$200
Field Lodgings	1 person x 186 days x \$12/day	\$1,860	\$372	May 2020	University of Oregon	\$2,232
Car Rental	1 car x 30 days x \$30/day	\$900	\$900	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$900
RNA later for sample collection	2 500ml bottles x \$454/bottle	\$908	\$908	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$908
Driver's Permit	1 person x 1 temporary driver's permit for Zanzibar	\$10	\$10	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$10
DNA Extraction	7 QIAGEN fecal DNA extraction kits x \$328/kit	\$2,300	\$2,300	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh	\$2,300

					Biodiversity Fund	
PCR Reactions	6 QIAGEN Multiplex PCR kits x \$2500/kit	\$15,000	\$15,000	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$15,000
Genotyping	Genotyping at 15 loci for 5400 reactions x \$132/96 well plate	\$7,500	\$7,500	July 2020	Mohamed bin Zayed Species Conservation Fund, Margot Marsh Biodiversity Fund	\$7,500
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	TOTAL PROJECT BUDGET (round to nearest whole dollar)	\$30,878	\$33,890			\$35,750

BUDGET JUSTIFICATION

State why each item listed in the CGF Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over \$500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

\$5k of the proposed lab and field work is being funded by U. of Oregon, with additional funds being sought from the Mohamed bin Zayed Species Conservation Fund (submitted), Margot Marsh Biodiversity Fund (submitted), National Geographic Society, and Primate Conservation Inc. The full project budget is laid out in this proposal, as we have not yet received notification about funding from any other agencies. Because of this, there is overlap in the requests we have made from other agencies and the AZA CGF. The intent is to combine any financial support received from the AZA CGF with funds from other grants to meet the full budget for the project. As soon as notification of funding from other agencies is received, the PI will send an updated budget to the AZA CGF.

Funding from this grant will cover costs associated with a mix of field work, lab work, and analyses. Six months of field work will be conducted in Zanzibar in order to collect non-invasive genetic samples from Zanzibar red colobus as well as to conduct interviews with people and collect social science data. Air fare and a visitor's visa are required for the PI to get to and enter Zanzibar, and field lodgings (6 months) will be necessary once in country. Non-invasive genetic sampling requires RNAlater (2 bottles at \$454 bottle; roughly 6ml used per sample) to stabilize DNA until the samples can be exported back to the US for DNA extraction. In addition, field assistants will be hired to help locate colobus and collect samples (funded by U. of Oregon). National Park fees, car rental (4 weeks at \$30/day), and a temporary driver's permit are all budget items necessary to complete both the genetics and social science components of the proposed work. The remaining budget items are all associated with lab work, which will take place at the U. of Oregon. Extraction kits are required to extract DNA from the fecal samples. 300 samples will be processed in total, which requires 7 QIAGEN fecal DNA extraction kits (7 kits at \$328/kit). Genotyping at 15 loci per sample, with 6 multiplexed reactions and 3 repeats per sample, will require 6 QIAGEN Multiplex PCR kits (\$2,500/kit). The samples will then be genotyped using an AB 3730 capillary DNA sequencer at Oregon State University's Center for Genome Research and Biocomputing. 57 plates of genotyping (96 wells each) are needed for all of our samples (\$132/plate). The combined costs associated with the full project total roughly \$30,878.

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ADDITIONAL RELEVANT REFERENCES

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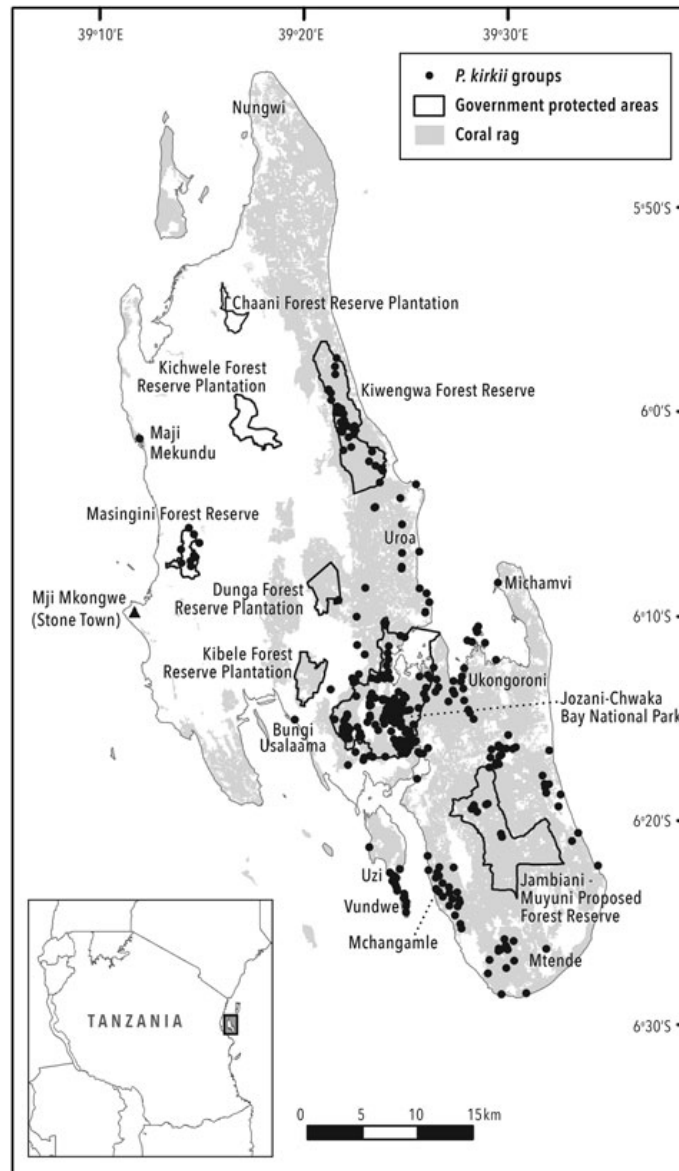
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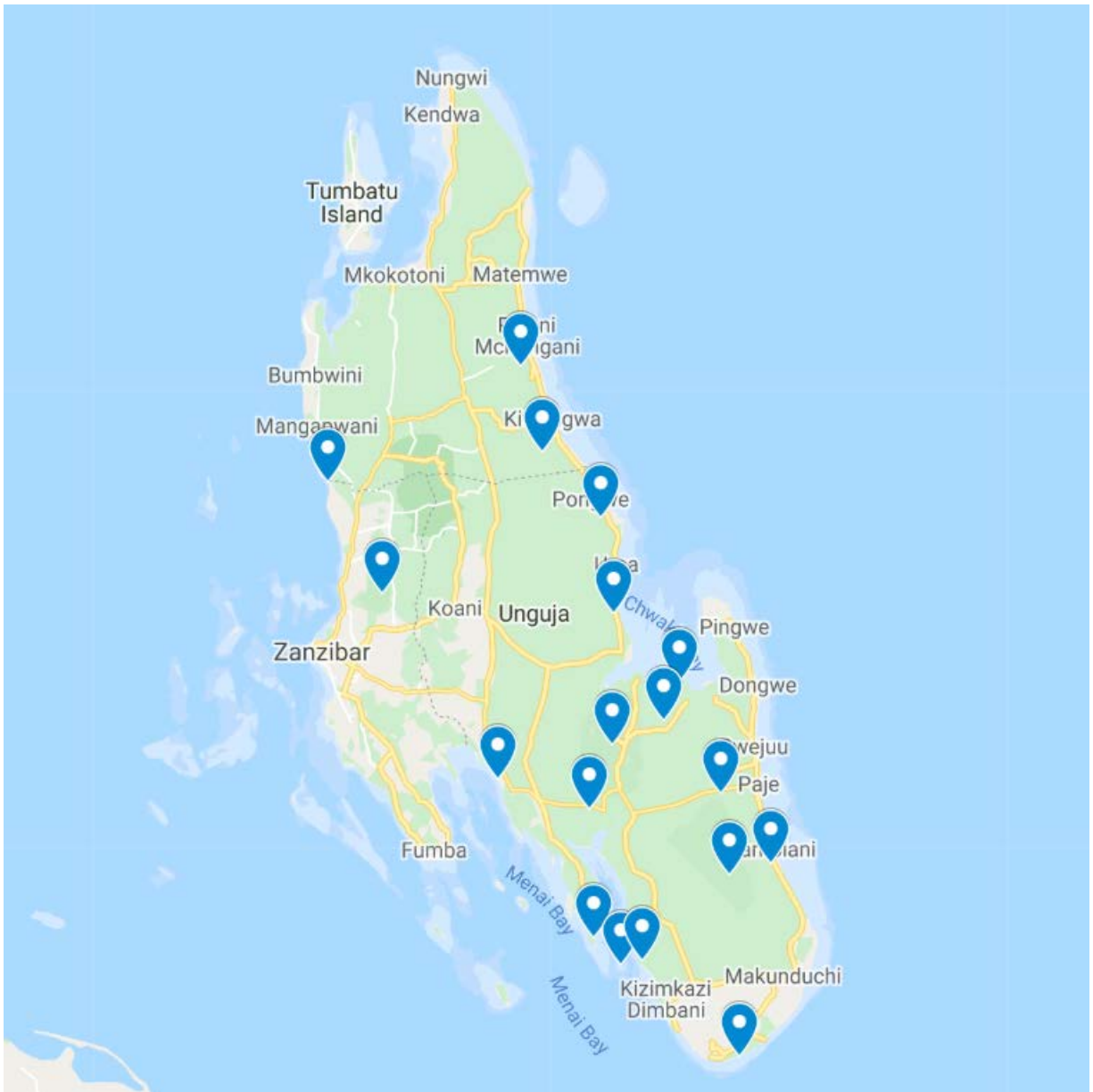
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Map 1: This project will take place on Unguja, the largest island of the Zanzibar Archipelago, United Republic of Tanzania, Africa.



Map 2: This map shows the locations of all Zanzibar red colobus (*Piliocolobus kirkii*) groups located and counted during a census carried out in 2015 on the island of Unguja, Zanzibar (Davenport et al. 2019).



Map 3: Non-invasive fecal sampling of Zanzibar red colobus will be carried out across Unguja. These sampling locales have been chosen based on Davenport et al.'s 2015 island-wide census of the Zanzibar red colobus population.

Tim R.B. Davenport PhD



Full name: Timothy Richard Bentley Davenport Nationality: British
Address: PO Box 2125, Arusha, TANZANIA tdavenport@wcs.org

Employment

2019 - present **Director - Species Conservation (Africa)**
Wildlife Conservation Society (WCS).

2006 - 2019 **Country Director - Tanzania**
Wildlife Conservation Society (WCS), PO Box 922, Zanzibar, **Tanzania**.

2004 - 2006 **Director - Southern Rift Program & SHCP**
Wildlife Conservation Society (WCS), PO Box 1475, Mbeya, **Tanzania. Malawi, Mozambique.**

2000 - 2008 **Director - Southern Highlands Conservation Program (SHCP)**
Wildlife Conservation Society (WCS). PO Box 1475, Mbeya, **Tanzania**.

1999 - 2000 **Chief Technical Advisor – Lake Rukwa Basin Integrated Project**
Wildlife Conservation Society (WCS). PO Box 1475, Mbeya, **Tanzania**.

1997 - 1998 **Project Manager - Lobéké Forest Project**
WorldWide Fund for Nature (WWF), BP 6776, Yaoundé, **Cameroon**.

1996 - 1997 **Biodiversity Consultant – EC & GEF Forest Cons. & Management Project**
The Forest Department, PO Box 1752, Kampala, **Uganda**.

1996 **Ecological Monitoring Consultant - Institute of Tropical Forest Conservation**
Mbarara University and Uganda National Parks c/o PO Box 7487, Kampala, **Uganda**.

(1994 – 1995) **Senior Local Consultant - GEF Sango Bay Biodiversity Project**
Makerere University Institute of Env. & Nat. Res., PO Box 7062, Kampala, **Uganda**.

1993 - 1996 **Field Programme Co-ordinator - EU & GEF Forest Conservation Project**
The Forest Department, PO Box 1752, Kampala, **Uganda**.

1991- 1992 **Biological Science Editor / Writer**
OCC, 213 Barns Road, Oxford, OX4 3UT, **UK**.

1986- 1990 **Postgraduate Research Assistant**
Department of Pure & Applied Biology, University of Leeds, Yorks, LS2 9JT, **UK**.

Other

2019 - pres Focal Point (East/Central Africa) – Key Biodiversity Areas (KBAs)

2019 - pres Member of IUCN Red Colobus Working Group

2012 - pres Member of IUCN Pangolin Specialist Group

2009 - pres Member of IUCN East Africa Primate Specialist Group

2007 - pres Member of IUCN Africa Herpetology Specialist Group

2012 - 2014 Chairman - Board of Governors, Zanzibar International School, Zanzibar

2002 - 2010 Honorary Consul - Southwest Tanzania - British Government. FCO.

Awards

2008 Awarded 'The 2008 Parker-Gentry International Award for Conservation' May 28th
Field Museum of Natural History, Chicago, USA.

2016 Nominated for the 2016 'Indianapolis Prize for Conservation'.

Dr Alexander Georgiev

Lecturer in Primatology | Director, Zanzibar
Red Colobus Project
School of Natural Sciences, Bangor University
Deiniol Road, Bangor, Gwynedd, LL57 2UW

E-mail: a.georgiev@bangor.ac.uk

<https://www.alexandergeorgiev.co.uk>

<https://www.zanzibarredcolobusproject.org>

EDUCATION

2012 Ph.D., Human Evolutionary Biology, Harvard University, USA
2008 A.M., Anthropology, Harvard University, USA
2005 B.Sc., Biology, Sofia University St. Kliment Ohridsky, Bulgaria

PROFESSIONAL APPOINTMENTS

2017 – Lecturer in Primatology, School of Natural Sciences, Bangor University, UK
2017 Lecturer (Winter Quarter 2017), Department of Anthropology, Northwestern University, USA
2015 – 2017 Postdoctoral Fellow, Laboratory for Human Biology Research, Department of Anthropology, Northwestern University.
2012 – 2015 Postdoctoral Scholar, Behavioral Biology Lab, and Fellow at the Institute for Mind and Biology, The University of Chicago.
2011 Teaching Assistant (Spring Term), Department of Human Evolutionary Biology, Harvard University
2008 – 2009 Teaching Assistant & Head Teaching Assistant, Harvard Extension School
2007 – 2009 Teaching Fellow, Department of Human Evolutionary Biology, Harvard University

PROFESSIONAL QUALIFICATIONS

2019 Associate Fellow of the Higher Education Academy, UK

SELECTED PUBLICATIONS

Olgun, H., Mohammed, M.K., Mzee, A.J., Green, M.E.L., Davenport, T.R.B. & **Georgiev, A.V.** (2020 & in review) The impact and implications of vehicle collisions on the endangered and endemic Zanzibar red colobus (*Piliocolobus kirkii*). bioRxiv doi: <https://doi.org/10.1101/2020.01.17.906909>

Georgiev, A.V., Melvin, Z.E., Warkentin, A-S., Winder, I.C., & Kassim, A. (2019) Two cases of dead-infant carrying by female Zanzibar red colobus (*Piliocolobus kirkii*) at Jozani-Chwaka Bay National Park, Zanzibar. *African Primates*, 13: 57-60.

Rosenfield, K.A, Semple, S., **Georgiev, A.V.**, Maestripieri, D., Higham, J.P. & Dubuc, C. (2019) Experimental evidence that female rhesus macaques (*Macaca mulatta*) perceive variation in male facial masculinity, *Royal Society Open Science*, 6: 181415

Kyweluk, M.A., **Georgiev, A.V.**, Borja, J.B., Gettler, L.T., & Kuzawa, C.K. (2018) Menarcheal timing is accelerated by favorable nutrition but unrelated to developmental cues of mortality or familial instability in Cebu, Philippines. *Evolution & Human Behavior*. 39: 76-81.

Milich, K.M., **Georgiev, A.V.**, Petersen, R.M., Emery Thompson, M. & Maestripieri, D. (2018) Alpha male status and availability of conceptive females are associated with high glucocorticoid concentrations in high-ranking male rhesus macaques (*Macaca mulatta*) during the mating season. *Hormones & Behavior*. 97: 5-13.

Petersdorf, M., Dubuc, C., **Georgiev, A.V.**, Winters, S., Higham, J.P. (2017) Is male rhesus macaque facial color under intra-sexual selection? *Behavioral Ecology*. 28: 1472-1481.

Georgiev, A.V., Ryan, C.P., Borja, J.B. & Kuzawa, C.W. (2017) Is early postnatal growth velocity, a proxy of minipubertal androgen action, related to adult second-to-fourth digit (2D:4D) ratios in men? A test in Cebu, Philippines. *American Journal of Human Biology*. 29: e23047

- Georgiev, A.V.**, Kuzawa, C.K. & McDade, T.W. (2016) Early developmental exposures shape trade-offs between acquired and innate immunity in humans. *Evolution, Medicine, and Public Health*. 256-269
doi:10.1093/emph/eow022
- McDade, T.W, **Georgiev, A.V.**, Kuzawa, C.W. (2016) Trade-offs in acquired and innate immune defenses in humans. *Evolution, Medicine, and Public Health*. doi: 10.1093/emph/eov033
- Georgiev, A.V.**, Christie, D., Rosenfield, K.A., Ruiz-Lambides, A.V., Maldonado. E., Emery Thompson, M. & Maestriperi, D. (2016) Breaking the succession rule: The costs and benefits of an alpha-status take-over by an immigrant rhesus macaque on Cayo Santiago. *Behaviour*. 153: 325-351
- Maestriperi, D. & **Georgiev, A.V.** (2016) What cortisol can tell us about the costs of sociality and reproduction among free-ranging rhesus macaque females on Cayo Santiago. Invited review. *American Journal of Primatology*. 78: 92-105
- Georgiev, A.V.**, Emery Thompson, M., Mandalaywala, T.M. & Maestriperi, D. (2015) Oxidative stress as an indicator of the costs of reproduction among free-ranging rhesus macaques. *Journal of Experimental Biology*. 218: 1981-1985
- Georgiev, A.V.**, Muehlenbein, M.P., Prall, S.P., Emery Thompson, M. & Maestriperi, D. (2015) Male quality, dominance rank and mating success in free-ranging rhesus macaques. *Behavioral Ecology*. 26: 763-772
- Georgiev, A.V.**, Russell, A.F., Emery Thompson, M., Otali, E., Muller, M.N. & Wrangham, R.W. (2014) The foraging costs of mating effort in male chimpanzees (*Pan troglodytes schweinfurthii*). *International Journal of Primatology*, 35, 725-745
- Emery Thompson, M. & **Georgiev, A.V.** (2014) The high price of success: costs of mating effort in male primates. *International Journal of Primatology*, 35, 609-627
- Georgiev, A.V.**, Klimczuk, A.C.E, Traficante, D.M. & Maestriperi, D. (2013) When violence pays: A cost-benefit analysis of aggressive behavior in animals and humans. *Evolutionary Psychology*, 11: 678-699
- Georgiev, A. V.**, Emery Thompson, M., Lokasola, A. L. & Wrangham, R. W. (2011) Seed predation by bonobos (*Pan paniscus*) at Kokolopori, Democratic Republic of Congo. *Primates*, 52: 309-314

EXTERNAL GRANTS

- 2017 Royal Society Research Grant: *The oxidative costs of reproduction in the endangered Zanzibar red colobus monkey* (£14,961). PI.
- 2015 International Primatological Society Research Grant: *Male oxidative stress and female mate choice in rhesus macaques* (\$1,500). PI.
- 2014 Leakey Foundation Research Grant: *Male oxidative stress and female mate choice in rhesus macaques* (\$24,996). PI.
- 2014 General Small Grant of the American Society of Primatologists: *Male oxidative stress and female mate choice in rhesus macaques* (\$1,200). PI.
- 2009 NSF Doctoral Dissertation Improvement Grant: *C-peptide of insulin as an index of energy expenditure in wild chimpanzees* (\$10,295). Co-PI (PI: Richard Wrangham).
- 2009 Wenner-Gren Foundation for Anthropological Research Dissertation Fieldwork Grant for research on male chimpanzee energetics and mating effort at Kanyawara, Kibale NP (\$20,813). Co-PI (PI: Richard Wrangham).
- 2009 International Primatological Society Research Grant for the study of chimpanzee energetics in Kibale NP (\$1,500).

ACADEMIC SERVICE

Member of the Research Working Party of the *Primate Society of Great Britain* (2017 -); Member of the Editorial Board of *American Journal of Primatology* (2018 -); Peer-review carried out for 19 academic journals and 4 research funding bodies to date.

JENNECA M. MCCARTER

University of Oregon, Department of Anthropology & Institute of Ecology and Evolution
302D Condon Hall, 1321 Kincaid St, Eugene, OR, 97403
jennecam@uoregon.edu

EDUCATION:

Ph.D. Student	2017 - present	University of Oregon Advisor: Dr. Nelson Ting	Anthropology
M.Sc.	2012	University of Cape Town Advisor: Dr. Jacqueline Bishop	Conservation Biology
B.A. Honors	2009	University of Arkansas Advisor: Dr. Cynthia Sagers	Biology

PROFESSIONAL EXPERIENCE:

2012 - 2017	Chimpanzee/Small Primate Zookeeper, Tulsa Zoo
2007 - 2011	Primate & Tropical American Rainforest Temporary Zookeeper, Tulsa Zoo, Tulsa, OK

RESEARCH EXPERIENCE

2011 - 2012	“Major Histocompatibility Complex Diversity in an Urban Chacma Baboon (<i>Papio ursinus</i>) Population: Implications for Conservation.” Research for MSc Thesis Western Cape Province, South Africa. Advisor: Jacqueline Bishop, University of Cape Town
2008	“Does the size-grain hypothesis explain allometric relationships in birds, bats, and beetles?” Research for Undergraduate Honors Thesis Punda Maria, Kruger National Park, Limpopo Province, South Africa Advisors: Laurence Kruger, Duke University and University of Cape Town. Mark Keith, University of the Witwatersrand. Cynthia Sagers, University of Arkansas
2008	“The effect of vegetation structure on spider and scorpion species diversity within a riparian zone in the Kruger National Park.” Skukuza, Kruger National Park, Mpumalanga Province, South Africa Advisors: Laurence Kruger, Duke University and University of Cape Town
2008	“Mimicry in Crab Spiders.” De Hoop Nature Reserve, Western Cape Town Province, South Africa. Advisor: Alan Ellis, Stellenbosch University

UNDERGRADUATE MENTOR EXPERIENCE

2010	Stevie Fitch (Tulane University), Lily Maynard (Smith College), Elizabeth Templeton (University of Arkansas) Research Project: “Does size matter? The role of habitat characteristics in defining the biodiversity of intertidal rock pools in the De Hoop Marine Protected Area.”
2010	Kirsten Gallaher (University of Cape Town), Eric Miner (Grinnell College), Julia Segel (Washington University) Research Project: “The importance of elephants and small mammals for Marula (<i>Sclerocarya birrea</i>) dispersal.”
2010	Hannah Cummins (Denison University), Sarah Lapidus (Amherst College), Evan Tasios (University of North Carolina)

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jennecam@uoregon.edu

- 2010 Research Project: “The association of bats with elephants in woodland savannas.”
Laura Budd (Middlebury College), Stevie Fitch (Tulane University), Sarah Lapidus (Amherst College), Clara Rowe (Amherst College)
Research Project: “A burning question: How do bark characteristics affect post-fire survival in three savanna tree species?”

AWARDS

- 2017 Promising Scholar Fellowship, University of Oregon
2008 Honors College Study Abroad Award, University of Arkansas
2008 Study Abroad Scholarship, Organization for Tropical Studies, Duke University
2005 Chancellor’s Award, University of Arkansas

SCIENCE OUTREACH

- 2017 – 2018 “Meet a Scientist Day”, Eugene Science Factory Children’s Museum, Eugene, OR
2018 - 2020 University of Oregon Science Open House, Molecular Anthropology Lab, University of Oregon, Eugene, OR
2012 – 2017 Chimpanzee Cooperative Feedings, Tulsa Zoo, Tulsa, OK
2007 - 2017 Primate Keeper Chats, Tulsa Zoo, Tulsa, OK
2012 “APE-ril (Great Ape Education Chats and Activities)”, Tulsa Zoo, Tulsa, OK

TEACHING EXPERIENCE

- 2017 - present Graduate Employee
University of Oregon, Eugene, OR
Courses Taught: Evolutionary Medicine, Human Origins, Evolution of Human Sexuality, Human Growth & Development, Human Biological Variation, Human Evolution

2010 Teaching Assistant
Organization for Tropical Studies/Duke University/University of the Witwatersrand
African Ecology & Conservation Study Abroad Semester/Summer REU, South Africa

CONFERENCES/WORKSHOPS

- 2018 41st Meeting of the American Society of Primatologists, San Antonio, TX

2016 Understanding Chimpanzees Symposium IV: Chimpanzees in Context, Lincoln Park Zoo, Chicago, IL

2013 Chimpanzee Husbandry Workshop (AZA Chimpanzee Species Survival Plan), Houston Zoo, Houston, TX

2011 Baboon Management Workshop (City of Cape Town), University of Cape Town, Cape Town, South Africa

2010 8th Annual Savanna Science Network Meeting, Skukuza, Kruger National Park, Mpumalanga Province, South Africa

PROFESSIONAL MEMBERSHIPS

Society for Conservation Biology
American Society of Primatologists
Association of Zoos and Aquariums

Zoe Elizabeth Melvin

Address: School of Natural Sciences, Bangor University, Deiniol Road, Bangor, Gwynedd, LL57 2UW, UK

Email: zoe.melvin@bangor.ac.uk

Education

September 2018 – Present – Bangor University: PhD in Conservation Biology (NERC ENVISION DTP), Case partner WCS Tanzania

Project: Living with human disturbance: conservation physiology and genetics in a wild African primate

July 2016 – Cardiff University: First Class Honours in BSc Zoology with Professional Training Year

Awards:

2016 – Edith Sheppard Prize – Best Final Year student in the area of Zoology

2015 – The most deserving student entering final year Zoology

2015 – The Bioscience Prize, Cardiff Naturalists Society, best project in the 2nd Year Field Course module

2013 – Certificate of Merit, consistent performance at a 1st class level in first year

Relevant modules:

Animal Physiological Adaptations (**82%**), Ecosystem Processes (**91%**), Field Course, Borneo (**79%**), Genes and Development 1 (**82%**), Population Ecology (**82%**), Research techniques (**78%**), Advanced Topics in Animal Behaviour (**71%**), Bioinformatics (**75%**), Conservation Biology (**73%**)

Final year project: Microsatellite genetic diversity of Bojer's skink (*Gongylomorphus bojerii*) in Ilot Vacoas, Ile aux Fouquets and Ile de la Passe, Mauritius, supervised by Professor Michael Bruford (**79%**) in conjunction with the Durrell Wildlife Conservation Trust.

June 2012 – King Edward VI Five Ways School

A-Level: Biology (A), Geography (A*), Psychology (A*), Mathematics (A), Further Mathematics (B)

AS-Level: English Language (A), General Studies (A) **GCSEs:** 13 A* including Mathematics and English

Professional Experience

January 2019 – Present – Field manager and long-term data collection coordinator, Zanzibar Red Colobus Project, Zanzibar

- Provide training, support and supervision to a team of local research assistants
- Oversee the collection and curation of long-term data for the project across multiple colobus groups
- Liaison between Zanzibar national park authorities and community organisations
- Management of day-to day logistical issues for the project
- Editor of the Keeping Up With The Colobus blog for the project and co-runs the ZRCP twitter page

September 2017 – August 2018 – Research Assistant, Paleo Primate Project, Oxford University and Gorongosa National Park, Mozambique

- Analysis and coding of video-recorded behavioural data
- Represented the project at international conferences, e.g. the Baboon Symposium 2017, Göttingen, Germany and the International Seminar on Biodiversity and Evolution, Kyoto University, Japan.
- Habituated and collected behavioural data from a group of wild baboons and collecting faecal samples for genetic analysis

November 2017 – Genome and Behaviour Field Course, Wildlife Research Centre, Kyoto University, Japan

- Behavioural observation and faecal samples collection from wild Japanese macaques on Yakushima island
- Extraction and amplification of mitochondrial DNA from faecal samples
- Spatial genetic analysis of macaques on Yakushima island

August 2016 – August 2017 – Primate Release Coordinator, CERCOPAN, Nigeria

- Coordinated the rehabilitation and release of three primate species: putty-nosed monkeys, mona monkeys and red-capped mangabeys, into the Cross River National Park, Nigeria
- Tracked and following released monkeys to collect behavioural and GPS data
- Managed local staff and working closely with Cross River National Park authorities

September 2014 – August 2015 – Primate Research Assistant, Primate and Predator Project, South Africa

- Conducted dawn to dusk behavioural surveys of three primate species: chacma baboons, samango monkeys and vervet monkeys both independently and as a group
- Gained an awareness of sedation methods of wild animals for research purposes
- Carried out habitat analysis of primate home ranges

Teaching and Demonstrating

- 2019 - Present Field supervision of undergraduate and postgraduate students for the Zanzibar red colobus project including advising on study design, data collection and analysis as well as handling logistical issues
- 2018 - 2019 Paid demonstration for various practicals for final-year undergraduate courses in Primatology including behavioural data analysis and GIS, Bangor University
- Nov 2019 Invited talk for Fieldwork Planning session for postgraduates, Bangor University
- Sept 2019 Postgraduate teaching assistant for Bangor University's undergraduate field course to Kibale National Park, Uganda: co-leading a practical on behavioural sampling, leading a session on GPS usage and advising on student projects
- August 2018 Taught multiple lectures on primatology and population census techniques during Oxford University's Gorongosa Field School for undergraduate students and Gorongosa's master's course

Publications

- du Plessis, S.J.*, Howard-McCombe, J.*, **Melvin, Z.E.***, Sheppard, E.C.*, *et al.* Genetic diversity and cryptic population re-establishment: management implications for the Bojer's skink (*Gongylomorphus bojerii*). *Conservation Genetics* (2018).
- Georgiev, A.V., **Melvin, Z.E.**, Warkentin, A.-S., Winder, I.C., Kassim, A. Two cases of dead-infant carrying by female Zanzibar red colobus (*Piliocolobus kirkii*) at Jozani-Chwaka Bay National Park, Zanzibar. *African Primates* (2019).
- In preparation (expected in 2020):
- Langhammer, P., Brooks, T., Bennett, C., Bruford, M.W., [and 22 others, including **Melvin, Z.**] (in prep.) The real impact of conservation action
- Melvin, Z.E.**, Blount, J., Georgiev, A.V. (in prep.) Oxidative stress and primatology. *International Journal of Primatology* (invited review)
- Melvin, Z.E., Georgiev, A.V. et al.** (in prep.) Male reproductive effort and oxidative stress in rhesus macaques.

Posters and Oral Presentations

- Zoe Melvin**, "Conservation Physiology of an Endangered Primate", PopScience seminar series for first-year undergraduates in biology and zoology, Bangor University, November 2019.
- Zoe Melvin**, Sandra Winters, James P. Higham, Dario Maestripieri, Melissa Emery Thompson, Alex Georgiev. "Facial colouration in male rhesus macaques does not reflect their ability to cope with oxidative stress",
- Poster presentation at Wildlife Research and Conservation Meeting, Berlin, October 2019.
 - Poster presentation at Ecology and Behaviour, Toulouse, May 2019.
 - Oral presentation at Association for the Study of Animal Behaviour meeting, York, April 2019.
- Zoe Melvin**, Ali Kassim, Isabelle Winder, Ann-Sophie Warkentin, Alexander Georgiev. "Prolonged carrying of dead infants in Zanzibar red colobus monkeys", poster presentation at the African Primatological Society Meeting, Entebbe, Uganda, September 2019.
- Zoe Melvin**, "Life as a primatologist", invited talk for the evening seminar series during the Bangor Primatology and Conservation Field Course, Kibale National Park, Uganda, September 2019.
- Zoe Melvin**, "A wee bit of stress: the conservation physiology of an endangered primate", invited seminar at the Behavioural Ecology and Physiology seminar series, Bangor University, April 2019.
- Zoe Melvin**, "Life as a primatologist", PopScience seminar series for first-year undergraduates in biology and zoology, Bangor University, December 2018.
- Zoe Melvin**, "Socio-ecological characterisation of the putative primate hybrid zone between *Papio cynocephalus* and *Papio ursinus* at Gorongosa, Mozambique," International Seminar on Biodiversity and Evolution, Kyoto University, November 2017.

Research Grants

- 2019 International Primatological Society** "Habitat disturbance and its effect on infant developmental rates and physiology in the Zanzibar red colobus (*Piliocolobus kirkii*)", \$2000 (\$500 for Conservation through Community Involvement)
- 2019 Primate Conservation Inc** "Habitat disturbance and its effect on infant developmental rates and physiology in the Zanzibar red colobus (*Piliocolobus kirkii*)", \$1725

Professional memberships

- Primatological Society of Great Britain, International Primatological Society, American Society of Primatologists, Genetics Society, Association for the Study of Animal Behaviour
- Member of the Education and Awareness Focus Group for the Red Colobus Conservation Network

Language competencies

- Fluent English, conversational Swahili

Nelson Ting – Curriculum Vitae

Department of Anthropology and Institute of Ecology and Evolution
308 Condon Hall
1218 University of Oregon
Eugene, OR 97403
nting@uoregon.edu

Higher education

- 2008 Ph.D., Anthropology, City University of New York (CUNY) Graduate Center
New York Consortium in Evolutionary Primatology (NYCEP)
Dissertation: Molecular Systematics of Red Colobus Monkeys (*Procolobus*
[*Piliocolobus*]): Understanding the Evolution of an Endangered Primate (Chair: E.
Delson)
- 2001 M.A., Anthropology, University of Missouri, Columbia
- 1999 B.A., Biology; Anthropology, Washington University, St. Louis

Professional and Academic Appointments

- 2014-present Associate Professor (with tenure), Department of Anthropology, Institute of Ecology and Evolution, Environmental Studies Program, African Studies Program, University of Oregon
- 2011 – 2014 Assistant Professor, Department of Anthropology, Institute of Ecology and Evolution, Environmental Studies Program, African Studies Program, University of Oregon
- 2008 – 2011 Assistant Professor, Department of Anthropology, University of Iowa

Relevant Publications

- Accepted Brand, C.M., M. Johnson, M.J. Ruiz-Lopez, L. Korte, A. Alonso, J.E. Maldonado, **N. Ting**. Abundance, density, and social structure of African forest elephants (*Loxodonta cyclotis*) in a human-modified landscape in southwestern Gabon. *PLoS One*
- Accepted Wikberg, E.C., D.M. Christie, P. Sicotte, **N. Ting**. Social interactions across groups of colobus monkeys (*Colobus vellerosus*) explain similarities in their gut microbiomes. *Animal Behavior*.
- In Press Linder, J.M., D. Cronin, **N. Ting**. Red Colobus (*Piliocolobus*) Conservation Action Plan. Primate Specialist Group.
- 2019 Simons, N.D., G. Eick, M.J. Ruiz-Lopez, D. Hyeroba, P.A. Omeja, G. Weny, C.A. Chapman, T.L. Goldberg, H.Q. Zheng, A. Shankar, W.M. Switzer, K.N. Sterner, **N. Ting**. Genome-wide patterns of gene expression in a wild primate indicate species-specific mechanisms associated with tolerance to natural simian immunodeficiency virus infection. *Genome Biology and Evolution*. 11(6): 1630-1643.

- 2019 Goodfellow, C.K, Wikberg, E.C, Christie, D.M., Sicotte, P., **N. Ting**. Divergence in gut microbial communities mirrors social group fission event in a black-and-white colobus monkey (*Colobus vellerosus*). *American Journal of Primatology*. e22966.
- 2016 Brand, C.M, F.J. White, M.T. Waller, M. L. Wakefield, M. J. Ruiz-Lopez, **N. Ting**. Initiation of genetic demographic monitoring of the bonobos (*Pan paniscus*) at Iyema, Lomako Forest, DRC. *Primate Conservation*. 30: 103-111.
- 2016 Ruiz-Lopez, M.J., C. Barelli, F. Rovero, K. Hodges, C. Roos, W.E. Peterman, **N. Ting**. A novel landscape genetics approach demonstrates the effects of human disturbance on the Udzungwa red colobus monkey (*Procolobus gordonorum*). *Heredity*. 116: 167-176. (Cover article)
- 2015 Oates, J.F. and **N. Ting**. Conservation consequences of unstable taxonomies: the case of the red colobus monkeys. In A.M. Behie and M.F. Oxnard (eds.). *Taxonomic Tapestries: The Threads of Evolutions Behavioural and Conservation Research*. Australia National University Press. Canberra, Australia.
- 2012 Wikberg, E.C., P. Sicotte, F.A. Campos, **N. Ting**. Variation in female dispersal, kin composition of groups, and proximity patterns in the white-thighed colobus monkey (*Colobus vellerosus*). *PLoS ONE*. 7(11): e48740.
- 2012 **Ting, N.**, C. Astaras, G. Hearn, S. Honarvar, J. Corush, A.S. Burrell, N. Phillips, B.J. Morgan, E.L. Gadsby, R.L. Raaum, C. Roos. Genetic signatures of a demographic collapse in a large-bodied forest dwelling primate (*Mandrillus leucophaeus*). *Ecology and Evolution*. 2: 550-561. (Cover article).

Recent Funding

- 2019 National Science Foundation, Biological Anthropology Program, Doctoral Dissertation Improvement Grant: Doctoral Dissertation Research: Social environment and gut microbiome plasticity in a black-and-white colobus monkey (*Colobus vellerosus*), **\$25,159 awarded**, PI: N. Ting, Co-PIs: Diana Christie, Brendan Bohannan.
- 2018 National Geographic Society Requests for Proposals: Genetic identity of elephant migrants across a forest-savanna mosaic of protected areas in Western Uganda, **\$75,000 awarded**, PI: N. Ting.

Professional Service

- 2019-present Scientific Advisor, IUCN Red Colobus Working Group
- 2018-present Associate Editor, Journal of Human Evolution
- 2013-present Member of the International Union for the Conservation of Nature (IUCN) Species Survival Commission (SSC) Primate Specialist Group, Africa Section



Association of Zoos & Aquariums

2020 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled "**SIS Institution Name**" for *each* institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)* or Institutional Review Board (IRB)** approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Bangor University has reviewed the proposal titled (*Proposal Title*) Intersecting human land-use, livelihoods, and conservation of the endangered Zanzibar red colobus and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Name S A THOMAS

Title CONTRACTS MANAGER

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

3/13/2020

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

* Institutional Animal Care and Use Committee (IACUC, for U.S. applicants) or the equivalent oversight committee (for non-U.S. applicants) approval is desirable for any proposal, but is required for any project in which animals are to be involved in invasive procedures and/or procedures that may result in pain, harm, distress, and/or altering of behavior (this includes any time an animal in the wild is sedated or anesthetized). This generally excludes samples obtained during regularly scheduled veterinary or husbandry procedures or emergency medical procedures. IACUC (or the international equivalent thereof) approval is required for both *in situ* and *ex situ* projects. Each institution involved in a project must submit the project proposal to its IACUC (or equivalent) for review. In order to be considered complete, proposals must include the Statement of Institutional Support from each participating institution/organization as recognition of compliance with these IACUC (or equivalent) approval guidelines.

If an institution is not registered as a research facility or similar facility and therefore does not have an IACUC (or equivalent), it needs to become affiliated with a university, zoo, or other body that has an IACUC (or equivalent) in order to obtain approval.

** Any projects supported through CGF funding that involve human subjects must be conducted under the auspices of a federally registered Institutional Review Board (IRB) and be consistent with 45 CFR Part 690: Federal Policy for the Protection of Human Subjects if the work or researchers are based in the USA. For members submitting from outside the USA for projects to be completed entirely outside the USA and its territories may use a process that meets the requirements of the prevailing legislation for protection of human subjects where the principal investigator is based or in the country where data collection will occur, whichever is more stringent. We note that all protocols must still conform to the types of protections outlined in 45 CFR Part 690 irrespective of the reviewing organization.

2020 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled “**SIS Institution Name**” for *each* institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)* or Institutional Review Board (IRB)** approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) University of Oregon has reviewed the proposal titled (*Proposal Title*) Intersecting human land-use, livelihoods, and conservation of the endangered Zanzibar red colobus and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Ashley Boehme

Name

Sponsored Projects Administrator

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

3/13/2020

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

* Institutional Animal Care and Use Committee (IACUC, for U.S. applicants) or the equivalent oversight committee (for non-U.S. applicants) approval is desirable for any proposal, but is required for any project in which animals are to be involved in invasive procedures and/or procedures that may result in pain, harm, distress, and/or altering of behavior (this includes any time an animal in the wild is sedated or anesthetized). This generally excludes samples obtained during regularly scheduled veterinary or husbandry procedures or emergency medical procedures. IACUC (or the international equivalent thereof) approval is required for both *in situ* and *ex situ* projects. Each institution involved in a project must submit the project proposal to its IACUC (or equivalent) for review. In order to be considered complete, proposals must include the Statement of Institutional Support from each participating institution/organization as recognition of compliance with these IACUC (or equivalent) approval guidelines.

If an institution is not registered as a research facility or similar facility and therefore does not have an IACUC (or equivalent), it needs to become affiliated with a university, zoo, or other body that has an IACUC (or equivalent) in order to obtain approval.

** Any projects supported through CGF funding that involve human subjects must be conducted under the auspices of a federally registered Institutional Review Board (IRB) and be consistent with 45 CFR Part 690: Federal Policy for the Protection of Human Subjects if the work or researchers are based in the USA. For members submitting from outside the USA for projects to be completed entirely outside the USA and its territories may use a process that meets the requirements of the prevailing legislation for protection of human subjects where the principal investigator is based or in the country where data collection will occur, whichever is more stringent. We note that all protocols must still conform to the types of protections outlined in 45 CFR Part 690 irrespective of the reviewing organization.

2020 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled “**SIS Institution Name**” for each institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)* or Institutional Review Board (IRB)** approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Wildlife Conservation Society (WCS) has reviewed the proposal titled (*Proposal Title*) _____ and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Dr Tim Davenport

Name

Director - Species Conservation (Africa)

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

14th March 2020

Date

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